

# To protect and serve

Agent 001 goes cabinet and power supply hunting, specifically for HTPC and high-end applications

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A childhood friend of mine recently upgraded to a GeForce GTX 260 graphics card after much persuasion. When we opened his cabinet to install it, I was horrified at the signs of rust. Then we discovered his system would not boot after adding this card. I immediately suspected the power supply, especially when I noticed it was a non-branded 400 watt unit and his older GeForce 7300GS would work fine. Thus began the hunt for a new cabinet and power supply.

Since HTPCs are super hot, HTPC cabinets are also the *in* thing and most vendors will be quick to point you to one. I recommend these cabinets with caution – sure they're compact but the cheaper ones are tacky and the cooling system woefully inadequate, unless you're building something based on the Atom platform. The attraction is the diminutive size, which some people will kill for – an unobtrusive cabinet to be tucked away behind furniture with a 42-inch LCD TV getting all the glory is a much sought after scenario. With shopping for HTPC cabinets do consider the better brands – their cabinets are built better and smarter (by the latter I refer to better cooling), and generally expansion support is better. Brands like CoolerMaster and Antec have some good models albeit pricier than other offerings in the market. Check out the Antec Fusion Remote Black, priced at Rs. 10,400. For the price you get a fully functional remote unit that works with the cabinets built in LCD display for a hands-free HTPC experience. The cabinet is built very well and cooled via two 120mm fans. An Antec 430-watt power supply available for Rs. 4,100 would be an ideal companion. Not only is it rated at above 80 percent in terms of efficiency but it's also very suitable for high-end HTPC applications. CoolerMasters Media 260 is also available for Rs. 8,100 and this is a really spacious case but available without power supply. Corsair's VX400 is a good PSU for this case available for Rs. 3,800. If you're looking for an ultra cheap HTPC chassis look at the Zebronic's Cube, available without a power supply for Rs. 2,500 – it's obviously

not as feature rich as the others mentioned above but for some affordable cannot be beaten.

For those looking at high-end rigs for gaming or generally powerful PCs we recommend larger, tower-type cabinets. By high-end I mean a quad core processor, 2-3 hard drives, 4 GB of RAM and a graphics card in the range of Rs. 15,000 or above. You can opt for a steel based chassis or one made of aluminium. The latter resists corrosion and that is why I recommend it. Steel cases without proper treatment tend to rust around the edges and this gradually worsens, especially if you live in a coastal region or any area with higher humidity. Cooling is also important; just having four and five fans isn't sufficient. The case should have good airflow. Sometimes wrongly placed fans can create cross currents of air, which is not good for cooling. You want the flow of air to be regular and strong and in general cool air should be sucked into the case from the front and possibly the sides and blown out through the rear. A couple of fans on the top of the case do not harm since they assist the process of exhausting air out. Cabinets that create a turbulence of air inside them are no good for cooling. Fans are of equal importance and most manufacturers offer very cheap fans with their cases. For someone blowing a lakh on a souped up gaming rig I recommend buying your own fans. Look at brands like

CoolerMaster, Antec, Scythe and NMB.

Larger fans are better for although their rpm is lower than a smaller fan they will actually move more air and silently owing to the larger diameter of the fan blade. The figure to look at when shopping for fans is their CFM rating indicating the cubic feet of air they can move in a minute, the higher this rating the better the fan. Of course the higher the rpm the noisier the fan – keep in mind.

If you really want something that is big, looks good and built of aluminium, therefore light and resisting corrosion look at the CoolerMaster ATCS 840 – a superlative case with very good cooling (it actually has 3 x 230 mm fans!). Priced at Rs. 15,000 the ATCS 840 represents the pinnacle of a high-end cabinet and supports liquid cooling solutions and up to 6 hard drives with no problems whatsoever. The intake fans also have dust filters – goodbye to regular cleaning. The Antec 1200 is priced at Rs. 13,000 and is another superb gaming chassis albeit not built of aluminium and without dust filters for the fans. This may be an issue for someone living in a dusty area, but in general I advocate the use of dust filters since even regular cleaning does not guarantee protection from a gradual rise in temperatures. This is because dust slowly works its wicked magic on fans inside the case and on components reducing their speed by gradually eroding their bearings. If you must buy something good and costly make sure it has dust filters since intake fans attract a lot of dust.

If you are going for a powerful dual GPU system like something based on a GTX 260 SLI, I recommend Corsairs HX 750 priced at Rs. 8,500. For single high-end graphics cards opt for Corsairs TX 650 priced at Rs. 6,500.

For a basic system I recommend a cabinet like CoolerMasters Elite 330 priced at Rs. 2,500. If you want something a little more powerful in terms of cooling and of course something that looks better, CoolerMasters 690 is also available for Rs. 4,000. A good power supply for any generic PC is Corsair's VX 450 – priced at Rs. 3,800. If you want to really squeeze it opt for a VIP Gold 400 watt, priced at Rs. 2,100. I don't recommend cheaper power supplies.. 

